

Bateria PF AC — Comparação (14)–(19)

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1. Tabela Resumo Global de Convergência e Clusters

Status abreviados: **OK**=LOCALLY_SOLVED · **INF**=INFEASIBLE/LOCALLY_INFEASIBLE · **ITL**=ITERATION_LIMIT · **TIM**=TIME_LIMIT · **KIL**=TIMEOUT_KILLED · **CRA**=SUBPROCESS_CRASH · **INV**=INVALID_MODEL · **ERR**=OTHER_ERROR

Grupos equivalentes: formulações dentro do mesmo {...} produziram resultado numericamente idêntico ($|\Delta|$ 1e-4 em vm/va/pg/qg/pf/qf). Grupos separados por · indicam soluções distintas.

Caso	(14)	(15)	(16)	(17)	(18)	(19)	Grupos equivalentes	Observações
01 MAXIMA NOTURNA_DEZ25.PWF	KIL	INF	KIL	INF	KIL	INF	—	Nenhuma formulação convergiu
300bus.pwf	INV	OK	OK	OK	OK	OK	{15} · {16} · {17} · {18} · {19}	não convergiram
3bus.pwf	OK	OK	OK	OK	OK	OK	{14} · {15,16} · {17,18,19}	
3bus_DBSH.pwf	OK	OK	OK	OK	OK	OK	{14} · {15,16} · {17,18,19}	
3bus_DCER.pwf	OK	OK	OK	OK	OK	OK	{14} · {15,16} · {17,18,19}	
3bus_DCSC.pwf	OK	OK	OK	OK	OK	OK	{14} · {15} · {16} · {17,18,19}	
3bus_DCline.pwf	INF	OK	OK	OK	OK	OK	{14} · {16,17,18,19}	não convergiram
3bus_DSHL.pwf	INF	OK	OK	OK	OK	OK	{15} · {16} · {17,18,19}	não convergiram
3bus_correcção.pwf	ERR	ERR	ERR	ERR	ERR	ERR	—	Nenhuma formulação convergiu
3bus_shunt_fields.pwf	OK	OK	OK	OK	OK	OK	{14} · {15} · {16} · {17,18,19}	
3busfrank.pwf	OK	OK	OK	OK	OK	OK	{14} · {15,16} · {17,18,19}	
3busfrank_continuous.pwf	OK	OK	OK	OK	OK	OK	{14} · {15,16} · {17,18,19}	
3busfrank_qdr.pwf	OK	OK	OK	OK	OK	OK	{14} · {15} · {16} · {17,18,19}	
4busfrank_vlf.pwf	INF	OK	OK	OK	OK	OK	{16} · {17,18,19}	não convergiram
500bus.pwf	INF	OK	OK	OK	OK	OK	{15} · {16} · {17,18} · {19}	não convergiram
5busfrank.pwf	OK	OK	OK	OK	OK	OK	{14} · {15,16} · {17,18,19}	
5busfrank_cns.pwf	OK	OK	OK	OK	OK	OK	{14} · {15,16} · {17,18,19}	
5busfrank_cse.pwf	INF	OK	OK	OK	OK	OK	{16} · {17,18,19}	não convergiram
5busfrank_cnf.pwf	OK	OK	OK	OK	OK	OK	{14} · {15,16} · {17} · {18,19}	
5busfrank_cap.pwf	OK	OK	OK	OK	OK	OK	{14} · {15,16} · {17} · {18,19}	
9bus.pwf	OK	OK	OK	OK	OK	OK	{14} · {15,16} · {17,18,19}	
9bus_transformer_fields.pwf	OK	OK	OK	OK	OK	OK	{14} · {15,16} · {17} · {18,19}	

Caso	(14)	(15)	(16)	(17)	(18)	(19)	Grupos equiva- lentes	Observações
test_default.pwf	ERR	ERR	ERR	ERR	ERR	ERR	—	Nenhuma formu- lação convergiu
test_line_short.pwf	ERR	ERR	ERR	ERR	ERR	ERR	—	Nenhuma formu- lação convergiu
test_system.pwf	ERR	ERR	OK	OK	OK	OK	{16} · {17,18,19}	nenhuma não convergi- ram

2. Análise Barra-a-Barra

Critério: para cada barra, listada apenas se vm ou va em (16)/(17)/(18)/(19) excede 0.0001 em relação a AMBAS (14) e (15). Cada barra divergente é exibida com uma linha para (14) e (15) (quando convergiram) e uma linha para cada script (16)–(19) que divergiu, permitindo comparar os valores absolutos lado a lado. Scripts (16)–(19) numericamente equivalentes (mesmo cluster) a (14) ou (15) são omitidos do caso. As colunas **pg** e **qg** somam a geração ativa e reativa (em pu) de todas as máquinas conectadas à barra; quando a barra não tem gerador, exibe-se **—**.

Caso 01 MAXIMA NOTURNA_DEZ25.PWF

Nenhuma referência (14) ou (15) convergiu para este caso; análise pulada.

Caso 300bus.pwf

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
1	(15)	1.0236	9.6571e-01	—	—
1	(16)	1.0704	9.2550e-01	—	—
1	(17)	1.0728	9.1794e-01	—	—
1	(18)	1.0659	8.5953e-01	—	—
1	(19)	1.0659	8.5822e-01	—	—
2	(15)	1.0323	9.8049e-01	—	—
2	(16)	1.0752	9.4032e-01	—	—
2	(17)	1.0773	9.3289e-01	—	—
2	(18)	1.0602	8.7885e-01	—	—
2	(19)	1.0602	8.7760e-01	—	—
3	(15)	0.9834	9.8003e-01	—	—
3	(16)	1.0215	9.3901e-01	—	—
3	(17)	1.0264	9.3123e-01	—	—
3	(18)	1.0105	8.7372e-01	—	—
3	(19)	1.0112	8.7240e-01	—	—
4	(15)	1.0073	9.1922e-01	—	—
4	(16)	1.0472	8.8319e-01	—	—
4	(17)	1.0523	8.7601e-01	—	—
4	(18)	1.1000	8.2244e-01	—	—
4	(19)	1.1000	8.2123e-01	—	—
5	(15)	1.0138	9.4322e-01	—	—
5	(16)	1.0606	9.0497e-01	—	—
5	(17)	1.0634	8.9747e-01	—	—
5	(18)	1.0549	8.3893e-01	—	—

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
5	(19)	1.0548	8.3763e-01	—	—
6	(15)	1.0269	9.7204e-01	—	—
6	(16)	1.0696	9.3237e-01	—	—
6	(17)	1.0723	9.2487e-01	—	—
6	(18)	1.0482	8.7054e-01	—	—
6	(19)	1.0483	8.6926e-01	—	—
7	(15)	0.9820	9.7762e-01	—	—
7	(16)	1.0214	9.3665e-01	—	—
7	(17)	1.0261	9.2888e-01	—	—
7	(18)	1.0127	8.7173e-01	—	—
7	(19)	1.0133	8.7037e-01	—	—
8	(15)	1.0140	8.6197e-01	—	0.8541
8	(16)	1.0463	8.3341e-01	—	0.1000
8	(17)	1.0492	8.2669e-01	—	0.1000
8	(18)	1.0443	7.7185e-01	—	0.1000
8	(19)	1.0436	7.7077e-01	—	0.1000
9	(15)	1.0001	9.0320e-01	—	—
9	(16)	1.0480	8.6807e-01	—	—
9	(17)	1.0509	8.6085e-01	—	—
9	(18)	1.0369	8.0156e-01	—	—
9	(19)	1.0363	8.0038e-01	—	—
10	(15)	1.0190	8.8075e-01	-0.0000	0.3253
10	(16)	1.0602	8.4691e-01	—	0.2000
10	(17)	1.0642	8.3992e-01	—	0.2000
10	(18)	1.1000	7.8625e-01	—	0.2000
10	(19)	1.1000	7.8493e-01	—	0.2000
11	(15)	1.0046	8.8898e-01	—	—
11	(16)	1.0525	8.5478e-01	—	—
11	(17)	1.0555	8.4772e-01	—	—
11	(18)	1.0367	7.8808e-01	—	—
11	(19)	1.0356	7.8704e-01	—	—
12	(15)	0.9895	9.5275e-01	—	—
12	(16)	1.0310	9.1314e-01	—	—
12	(17)	1.0351	9.0564e-01	—	—
12	(18)	1.0286	8.4844e-01	—	—
12	(19)	1.0295	8.4712e-01	—	—
13	(15)	0.9963	8.3826e-01	—	—
13	(16)	1.0438	8.0817e-01	—	—
13	(17)	1.0469	8.0141e-01	—	—
13	(18)	0.9847	7.3947e-01	—	—
13	(19)	0.9843	7.3829e-01	—	—
14	(15)	0.9767	6.7894e-01	—	—
14	(16)	1.0092	6.6355e-01	—	—
14	(17)	1.0128	6.5837e-01	—	—
14	(18)	1.0373	6.1041e-01	—	—
14	(19)	1.0358	6.0942e-01	—	—
15	(15)	1.0049	5.5530e-01	—	—
15	(16)	1.0323	5.4923e-01	—	—
15	(17)	1.0361	5.4533e-01	—	—
15	(18)	1.0880	5.0806e-01	—	—
15	(19)	1.0856	5.0714e-01	—	—
16	(15)	0.9859	6.8236e-01	—	—
16	(16)	1.0242	6.6616e-01	—	—
16	(17)	1.0291	6.6135e-01	—	—
16	(18)	1.0528	6.2022e-01	—	—

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
16	(19)	1.0537	6.1957e-01	—	—
17	(15)	1.0521	4.6157e-01	—	—
17	(16)	1.0643	4.6920e-01	—	—
17	(17)	1.0683	4.6594e-01	—	—
17	(18)	1.1000	4.3346e-01	—	—
17	(19)	1.1000	4.3254e-01	—	—
19	(15)	0.9686	8.5525e-01	—	—
19	(16)	1.0164	8.2149e-01	—	—
19	(17)	1.0204	8.1467e-01	—	—
19	(18)	1.0354	7.5945e-01	—	—
19	(19)	1.0359	7.5829e-01	—	—
20	(15)	0.9990	8.0685e-01	—	0.5337
20	(16)	1.0455	7.7912e-01	—	0.2000
20	(17)	1.0484	7.7255e-01	—	0.2000
20	(18)	0.9444	7.0449e-01	—	0.2000
20	(19)	0.9446	7.0329e-01	—	0.2000
21	(15)	0.9683	8.7807e-01	—	—
21	(16)	1.0144	8.4368e-01	—	—
21	(17)	1.0175	8.3681e-01	—	—
21	(18)	1.0359	7.8013e-01	—	—
21	(19)	1.0365	7.7893e-01	—	—

(mostrando 20 de 300 barras divergentes; tabela truncada)

Caso 3bus.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
1	(14)	1.1000	-1.5046e-35	0.1557	0.0303
1	(15)	1.0290	-1.3749e-35	0.1829	-0.0052
1	(17)	1.0071	0.0000e+00	0.1833	0.0012
1	(18)	1.0071	0.0000e+00	0.1833	0.0012
1	(19)	1.0071	0.0000e+00	0.1833	0.0012
2	(14)	1.1000	-3.8931e-10	0.1557	0.0303
2	(15)	1.0300	-7.0907e-03	0.1299	0.0670
2	(17)	1.0073	-6.6894e-03	0.1299	0.0610
2	(18)	1.0073	-6.6894e-03	0.1299	0.0610
2	(19)	1.0073	-6.6894e-03	0.1299	0.0610
3	(14)	1.0698	-1.8371e-02	—	—
3	(15)	0.9970	-2.4609e-02	—	—
3	(17)	0.9740	-2.5383e-02	—	—
3	(18)	0.9740	-2.5383e-02	—	—
3	(19)	0.9740	-2.5383e-02	—	—

Caso 3bus_DBSH.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
1	(14)	0.9693	0.0000e+00	0.1660	0.2135
1	(15)	1.0290	6.4160e-38	0.2032	0.1904

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
1	(17)	1.0997	0.0000e+00	0.2041	0.2291
1	(18)	1.0997	0.0000e+00	0.2041	0.2291
1	(19)	1.0997	0.0000e+00	0.2041	0.2291
2	(14)	0.9693	-2.3056e-09	0.1660	0.2135
2	(15)	1.0300	-9.6402e-03	0.1299	0.2855
2	(17)	1.0997	-7.6171e-03	0.1299	0.3072
2	(18)	1.0997	-7.6171e-03	0.1299	0.3072
2	(19)	1.0997	-7.6143e-03	0.1299	0.3072
3	(14)	0.9000	1.1186e-02	—	—
3	(15)	0.9602	9.4349e-03	—	—
3	(17)	1.0300	1.3387e-02	—	—
3	(18)	1.0300	1.3387e-02	—	—
3	(19)	1.0300	1.3388e-02	—	—

Caso 3bus_DCER.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
1	(14)	0.9693	0.0000e+00	0.1660	0.2135
1	(15)	1.0290	6.4160e-38	0.2032	0.1904
1	(17)	1.0221	0.0000e+00	0.2043	0.1261
1	(18)	1.0221	0.0000e+00	0.2043	0.1261
1	(19)	1.0221	0.0000e+00	0.2043	0.1261
2	(14)	0.9693	-2.3056e-09	0.1660	0.2135
2	(15)	1.0300	-9.6402e-03	0.1299	0.2855
2	(17)	1.0300	-1.7178e-02	0.1299	0.3481
2	(18)	1.0300	-1.7178e-02	0.1299	0.3481
2	(19)	1.0300	-1.7178e-02	0.1299	0.3481
3	(14)	0.9000	1.1186e-02	—	—
3	(15)	0.9602	9.4349e-03	—	—
3	(17)	0.9567	5.4726e-03	—	—
3	(18)	0.9567	5.4726e-03	—	—
3	(19)	0.9567	5.4725e-03	—	—

Caso 3bus_DCSC.pwf

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
1	(14)	1.0998	4.4373e-38	0.0003	0.0142
1	(15)	1.0290	0.0000e+00	0.0498	0.0662
1	(16)	1.0352	0.0000e+00	0.0468	0.1756
1	(17)	1.0064	0.0000e+00	0.0470	-0.0055
1	(18)	1.0064	0.0000e+00	0.0470	-0.0055
1	(19)	1.0064	0.0000e+00	0.0470	-0.0055
2	(14)	1.0998	-1.1064e-10	0.0003	0.0142
2	(15)	1.0300	-2.3356e-04	0.1299	-0.2536
2	(16)	1.0362	-2.3972e-04	0.1299	-0.1460
2	(17)	1.0062	-2.6434e-04	0.1299	0.0365
2	(18)	1.0062	-2.6434e-04	0.1299	0.0365
2	(19)	1.0062	-2.6434e-04	0.1299	0.0365
3	(14)	1.0975	2.0890e-03	0.3035	0.0250
3	(15)	1.0300	-3.0841e-02	0.1299	0.2460

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
3	(16)	1.0178	-1.2142e-02	0.1299	0.0250
3	(17)	0.9878	-1.2786e-02	0.1299	0.0250
3	(18)	0.9878	-1.2786e-02	0.1299	0.0250
3	(19)	0.9878	-1.2786e-02	0.1299	0.0250

Caso 3bus_DCline.pwf

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
2	(14)	1.1000	1.6758e-07	0.0461	-2.8638
2	(16)	1.1000	1.7794e+00	0.1299	0.5556
2	(17)	1.1000	1.7794e+00	0.1299	0.5556
2	(18)	1.1000	1.7794e+00	0.1299	0.5556
2	(19)	1.1000	1.7794e+00	0.1299	0.5556
3	(14)	1.0698	-1.8371e-02	—	—
3	(16)	0.9000	1.8754e-01	—	—
3	(17)	0.9000	1.8754e-01	—	—
3	(18)	0.9000	1.8754e-01	—	—
3	(19)	0.9000	1.8754e-01	—	—

Caso 3bus_DSHL.pwf

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
1	(15)	1.0290	0.0000e+00	1.2045	3.1908
1	(16)	1.0458	0.0000e+00	0.3514	1.3040
1	(17)	1.0458	0.0000e+00	0.3514	1.3040
1	(18)	1.0458	0.0000e+00	0.3514	1.3040
1	(19)	1.0458	0.0000e+00	0.3514	1.3040
2	(15)	0.6622	3.3184e-01	—	—
2	(16)	0.9000	1.3356e-01	—	—
2	(17)	0.9000	1.3358e-01	—	—
2	(18)	0.9000	1.3358e-01	—	—
2	(19)	0.9000	1.3358e-01	—	—
3	(15)	0.6952	2.0551e-01	—	—
3	(16)	0.9231	5.1343e-02	—	—
3	(17)	0.9231	5.1315e-02	—	—
3	(18)	0.9231	5.1315e-02	—	—
3	(19)	0.9231	5.1315e-02	—	—

Caso 3bus_corrections.pwf

Nenhuma referência (14) ou (15) convergiu para este caso; análise pulada.

Caso 3bus_shunt_fields.pwf

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
1	(14)	0.9418	-1.8478e-36	0.1847	0.2627
1	(15)	1.0290	3.0075e-39	0.1844	-0.1078
1	(16)	1.0522	0.0000e+00	0.2002	0.3048
1	(17)	1.0397	0.0000e+00	0.1959	0.1085
1	(18)	1.0397	0.0000e+00	0.1959	0.1085

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
1	(19)	1.0397	0.0000e+00	0.1959	0.1085
2	(14)	0.9005	4.1604e-02	0.1445	—
2	(15)	1.0300	-7.1797e-03	0.1299	0.4192
2	(16)	1.0062	3.8981e-02	0.1299	—
2	(17)	1.0084	2.4423e-02	0.1299	—
2	(18)	1.0084	2.4423e-02	0.1299	—
2	(19)	1.0084	2.4423e-02	0.1299	—
3	(14)	0.9000	-2.3446e-02	—	—
3	(15)	1.0147	-4.2134e-02	—	—
3	(16)	1.0143	-1.9622e-02	—	—
3	(17)	1.0198	-3.7784e-02	—	—
3	(18)	1.0198	-3.7784e-02	—	—
3	(19)	1.0198	-3.7784e-02	—	—

Caso 3busfrank.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
1	(14)	1.1000	-4.5139e-36	0.2324	0.1241
1	(15)	1.0290	5.6424e-37	0.3412	0.0095
1	(17)	1.0189	0.0000e+00	0.3407	0.0715
1	(18)	1.0189	0.0000e+00	0.3407	0.0715
1	(19)	1.0189	0.0000e+00	0.3407	0.0715
2	(14)	1.1000	-2.6818e-17	0.2324	0.1241
2	(15)	1.0300	-2.5477e-02	0.1299	0.2447
2	(17)	1.0129	-1.8667e-02	0.1299	0.1822
2	(18)	1.0129	-1.8667e-02	0.1299	0.1822
2	(19)	1.0129	-1.8667e-02	0.1299	0.1822
3	(14)	1.0423	-1.5680e-02	—	—
3	(15)	0.9672	-3.0800e-02	—	—
3	(17)	0.9527	-2.7882e-02	—	—
3	(18)	0.9527	-2.7882e-02	—	—
3	(19)	0.9527	-2.7881e-02	—	—

Caso 3busfrank_continuous_shunt.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
1	(14)	1.1000	0.0000e+00	0.2301	0.2348
1	(15)	1.0290	-1.5742e-35	0.3330	0.0977
1	(17)	1.0247	0.0000e+00	0.3323	0.1971
1	(18)	1.0247	0.0000e+00	0.3323	0.1971
1	(19)	1.0247	0.0000e+00	0.3323	0.1971
2	(14)	1.1000	1.7965e-18	0.2301	0.2348
2	(15)	1.0300	-1.2754e-02	0.1299	0.3449
2	(17)	1.0200	-7.0006e-03	0.1299	0.2420
2	(18)	1.0200	-7.0006e-03	0.1299	0.2420
2	(19)	1.0200	-7.0006e-03	0.1299	0.2420
3	(14)	1.0625	1.0513e-03	—	—
3	(15)	0.9906	-6.4858e-03	—	—

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
3	(17)	0.9833	-3.7295e-03	—	—
3	(18)	0.9833	-3.7295e-03	—	—
3	(19)	0.9833	-3.7295e-03	—	—

Caso 3busfrank_qlim.pwf

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
1	(14)	1.1000	-3.7150e-37	0.3137	0.5609
1	(15)	1.0290	3.9120e-35	0.5341	0.3521
1	(16)	1.0526	0.0000e+00	0.5207	0.4358
1	(17)	1.0750	0.0000e+00	0.5156	0.6649
1	(18)	1.0750	0.0000e+00	0.5156	0.6649
1	(19)	1.0750	0.0000e+00	0.5156	0.6649
2	(14)	1.1000	4.8617e-08	0.3137	0.5609
2	(15)	1.0300	-5.0419e-02	0.1299	0.8048
2	(16)	1.0448	-3.7207e-02	0.1299	0.7040
2	(17)	1.0439	-1.0585e-02	0.1299	0.4742
2	(18)	1.0439	-1.0585e-02	0.1299	0.4742
2	(19)	1.0439	-1.0584e-02	0.1299	0.4742
3	(14)	0.9606	4.4430e-02	—	—
3	(15)	0.8761	2.6851e-02	—	—
3	(16)	0.9000	3.0818e-02	—	—
3	(17)	0.9126	4.3346e-02	—	—
3	(18)	0.9126	4.3346e-02	—	—
3	(19)	0.9126	4.3346e-02	—	—

Caso 4busfrank_vlim.pwf

Nenhuma referência (14) ou (15) convergiu para este caso; análise pulada.

Caso 500bus.pwf

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
1	(15)	1.0115	-1.4985e-01	—	—
1	(16)	1.0117	-1.5096e-01	—	—
1	(17)	1.0032	-1.5828e-01	—	—
1	(18)	1.0032	-1.5828e-01	—	—
1	(19)	1.0032	-1.5828e-01	—	—
2	(15)	1.0109	-1.5110e-01	—	—
2	(16)	1.0112	-1.5221e-01	—	—
2	(17)	1.0026	-1.5955e-01	—	—
2	(18)	1.0026	-1.5955e-01	—	—
2	(19)	1.0026	-1.5955e-01	—	—
3	(15)	1.0037	-1.7275e-01	—	—
3	(16)	1.0038	-1.7382e-01	—	—
3	(17)	1.0003	-1.8190e-01	—	—
3	(18)	1.0003	-1.8190e-01	—	—
3	(19)	1.0003	-1.8190e-01	—	—
4	(15)	1.0016	-1.7715e-01	—	—
4	(16)	1.0016	-1.7822e-01	—	—
4	(17)	0.9981	-1.8633e-01	—	—
4	(18)	0.9981	-1.8633e-01	—	—

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
4	(19)	0.9981	-1.8633e-01	—	—
5	(15)	1.0147	1.6051e-02	—	—
5	(16)	1.0145	1.5011e-02	—	—
5	(17)	0.9983	1.5141e-02	—	—
5	(18)	0.9983	1.5141e-02	—	—
5	(19)	0.9983	1.5141e-02	—	—
6	(15)	1.0146	1.5944e-02	—	—
6	(16)	1.0144	1.4904e-02	—	—
6	(17)	0.9983	1.5031e-02	—	—
6	(18)	0.9983	1.5031e-02	—	—
6	(19)	0.9983	1.5031e-02	—	—
7	(15)	1.0327	-8.2623e-02	—	—
7	(16)	1.0346	-8.3970e-02	—	—
7	(17)	1.0329	-9.0338e-02	—	—
7	(18)	1.0329	-9.0338e-02	—	—
7	(19)	1.0329	-9.0339e-02	—	—
8	(15)	1.0244	-1.1376e-01	—	—
8	(16)	1.0256	-1.1500e-01	—	—
8	(17)	1.0211	-1.2184e-01	—	—
8	(18)	1.0211	-1.2184e-01	—	—
8	(19)	1.0211	-1.2184e-01	—	—
9	(15)	1.0400	-3.3735e-02	7.7180	1.2322
9	(16)	1.0434	-3.5341e-02	7.7180	1.4643
9	(17)	1.0554	-4.2302e-02	7.7180	3.5970
9	(18)	1.0554	-4.2302e-02	7.7180	3.5970
9	(19)	1.0554	-4.2302e-02	7.7180	3.5970
10	(15)	1.0170	-2.0662e-01	—	—
10	(16)	1.0166	-2.0775e-01	—	—
10	(17)	0.9892	-2.1654e-01	—	—
10	(18)	0.9892	-2.1654e-01	—	—
10	(19)	0.9892	-2.1654e-01	—	—
11	(15)	1.0170	-2.0669e-01	—	—
11	(16)	1.0166	-2.0783e-01	—	—
11	(17)	0.9892	-2.1662e-01	—	—
11	(18)	0.9892	-2.1662e-01	—	—
11	(19)	0.9892	-2.1662e-01	—	—
12	(15)	0.9950	-2.2144e-01	—	—
12	(16)	0.9945	-2.2262e-01	—	—
12	(17)	0.9718	-2.3197e-01	—	—
12	(18)	0.9718	-2.3197e-01	—	—
12	(19)	0.9718	-2.3197e-01	—	—
13	(15)	0.9944	-2.2256e-01	—	—
13	(16)	0.9940	-2.2373e-01	—	—
13	(17)	0.9712	-2.3314e-01	—	—
13	(18)	0.9712	-2.3314e-01	—	—
13	(19)	0.9712	-2.3314e-01	—	—
14	(15)	1.0311	-4.3247e-02	—	—
14	(16)	1.0275	-4.3792e-02	—	—
14	(17)	1.0088	-4.5823e-02	—	—
14	(18)	1.0088	-4.5823e-02	—	—
14	(19)	1.0088	-4.5823e-02	—	—
15	(15)	1.0237	-8.5034e-02	—	—
15	(16)	1.0210	-8.5849e-02	—	—
15	(17)	1.0028	-8.9454e-02	—	—
15	(18)	1.0028	-8.9454e-02	—	—

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
15	(19)	1.0028	-8.9454e-02	—	—
16	(15)	1.0400	6.3568e-03	4.4440	0.7636
16	(16)	1.0356	6.2176e-03	4.4440	0.6916
16	(17)	1.0137	6.3037e-03	4.4440	0.4073
16	(18)	1.0137	6.3037e-03	4.4440	0.4073
16	(19)	1.0137	6.3036e-03	4.4440	0.4073
17	(15)	1.0400	0.0000e+00	6.4761	1.2915
17	(16)	1.0327	0.0000e+00	6.4768	0.7460
17	(17)	1.0123	0.0000e+00	6.5175	0.5002
17	(18)	1.0123	0.0000e+00	6.5175	0.5002
17	(19)	1.0123	0.0000e+00	6.5175	0.5002
18	(15)	1.0400	3.1740e-02	1.5760	0.1972
18	(16)	1.0390	3.1463e-02	1.5760	0.2489
18	(17)	1.0169	3.2590e-02	1.5760	0.1783
18	(18)	1.0169	3.2590e-02	1.5760	0.1783
18	(19)	1.0169	3.2590e-02	1.5760	0.1783
19	(15)	1.0178	-2.1213e-01	—	—
19	(16)	1.0174	-2.1324e-01	—	—
19	(17)	0.9856	-2.2232e-01	—	—
19	(18)	0.9856	-2.2232e-01	—	—
19	(19)	0.9856	-2.2232e-01	—	—
20	(15)	1.0178	-2.1229e-01	—	—
20	(16)	1.0173	-2.1341e-01	—	—
20	(17)	0.9855	-2.2250e-01	—	—
20	(18)	0.9855	-2.2250e-01	—	—
20	(19)	0.9855	-2.2250e-01	—	—

(mostrando 20 de 500 barras divergentes; tabela truncada)

Caso 5busfrank.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
1	(14)	1.1000	-4.5139e-36	0.2379	0.1489
1	(15)	1.0290	0.0000e+00	0.3532	0.0283
1	(17)	1.0378	0.0000e+00	0.3514	0.0879
1	(18)	1.0378	0.0000e+00	0.3514	0.0879
1	(19)	1.0378	0.0000e+00	0.3514	0.0879
2	(14)	1.1000	-4.6496e-17	0.2379	0.1489
2	(15)	1.0300	-2.6945e-02	0.1299	0.2764
2	(17)	1.0323	-1.9556e-02	0.1299	0.2151
2	(18)	1.0323	-1.9556e-02	0.1299	0.2151
2	(19)	1.0323	-1.9556e-02	0.1299	0.2151
3	(14)	1.0374	-1.2668e-02	—	—
3	(15)	0.9618	-2.8079e-02	—	—
3	(17)	0.9679	-2.4182e-02	—	—
3	(18)	0.9679	-2.4182e-02	—	—
3	(19)	0.9679	-2.4182e-02	—	—
4	(14)	1.0281	-6.0660e-03	—	—
4	(15)	0.9518	-2.0388e-02	—	—
4	(17)	0.9579	-1.6588e-02	—	—
4	(18)	0.9579	-1.6588e-02	—	—
4	(19)	0.9579	-1.6588e-02	—	—

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
5	(14)	1.0235	-2.7201e-03	—	—
5	(15)	0.9468	-1.6481e-02	—	—
5	(17)	0.9529	-1.2731e-02	—	—
5	(18)	0.9529	-1.2731e-02	—	—
5	(19)	0.9529	-1.2731e-02	—	—

Caso 5busfrank_cphs.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
2	(14)	1.1000	3.6956e-09	0.2923	0.1203
2	(15)	1.0300	-4.0105e-02	0.1299	0.3080
2	(17)	1.0403	-5.1113e-02	0.1299	0.4060
2	(18)	1.0403	-5.1113e-02	0.1299	0.4060
2	(19)	1.0403	-5.1114e-02	0.1299	0.4060
3	(14)	1.0332	-2.5593e-02	—	—
3	(15)	0.9570	-4.9592e-02	—	—
3	(17)	0.9625	-5.4916e-02	—	—
3	(18)	0.9625	-5.4916e-02	—	—
3	(19)	0.9625	-5.4916e-02	—	—
4	(14)	1.0060	-1.5136e-02	—	—
4	(15)	0.9275	-3.7346e-02	—	—
4	(17)	0.9331	-4.2813e-02	—	—
4	(18)	0.9331	-4.2813e-02	—	—
4	(19)	0.9331	-4.2813e-02	—	—
5	(14)	0.9923	-9.6918e-03	—	—
5	(15)	0.9126	-3.0925e-02	—	—
5	(17)	0.9183	-3.6470e-02	—	—
5	(18)	0.9183	-3.6470e-02	—	—
5	(19)	0.9183	-3.6471e-02	—	—

Caso 5busfrank_csca.pwf

Nenhuma referência (14) ou (15) convergiu para este caso; análise pulada.

Caso 5busfrank_ctaf.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
2	(14)	1.1000	3.6956e-09	0.2923	0.1203
2	(15)	1.0300	-4.0105e-02	0.1299	0.3080
2	(17)	1.0403	-5.1113e-02	0.1299	0.4060
2	(18)	1.0126	-2.1727e-02	0.1299	0.1486
2	(19)	1.0126	-2.1727e-02	0.1299	0.1486
3	(14)	1.0332	-2.5593e-02	—	—
3	(15)	0.9570	-4.9592e-02	—	—
3	(17)	0.9625	-5.4916e-02	—	—
3	(18)	0.9478	-4.0841e-02	—	—
3	(19)	0.9478	-4.0842e-02	—	—
4	(14)	1.0060	-1.5136e-02	—	—
4	(15)	0.9275	-3.7346e-02	—	—

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
4	(17)	0.9331	-4.2813e-02	—	—
4	(18)	0.9922	-3.0099e-02	—	—
4	(19)	0.9922	-3.0099e-02	—	—
5	(14)	0.9923	-9.6918e-03	—	—
5	(15)	0.9126	-3.0925e-02	—	—
5	(17)	0.9183	-3.6470e-02	—	—
5	(18)	0.9894	-2.4624e-02	—	—
5	(19)	0.9894	-2.4624e-02	—	—

Caso 5busfrank_ctap.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
2	(14)	1.1000	3.6956e-09	0.2923	0.1203
2	(15)	1.0300	-4.0105e-02	0.1299	0.3080
2	(17)	1.0403	-5.1113e-02	0.1299	0.4060
2	(18)	1.0126	-2.1727e-02	0.1299	0.1486
2	(19)	1.0126	-2.1727e-02	0.1299	0.1486
3	(14)	1.0332	-2.5593e-02	—	—
3	(15)	0.9570	-4.9592e-02	—	—
3	(17)	0.9625	-5.4916e-02	—	—
3	(18)	0.9478	-4.0841e-02	—	—
3	(19)	0.9478	-4.0842e-02	—	—
4	(14)	1.0060	-1.5136e-02	—	—
4	(15)	0.9275	-3.7346e-02	—	—
4	(17)	0.9331	-4.2813e-02	—	—
4	(18)	0.9922	-3.0099e-02	—	—
4	(19)	0.9922	-3.0099e-02	—	—
5	(14)	0.9923	-9.6918e-03	—	—
5	(15)	0.9126	-3.0925e-02	—	—
5	(17)	0.9183	-3.6470e-02	—	—
5	(18)	0.9894	-2.4624e-02	—	—
5	(19)	0.9894	-2.4624e-02	—	—

Caso 9bus.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
1	(14)	1.1000	-1.1456e-29	1.5765	0.1210
1	(15)	1.0750	0.0000e+00	1.4249	0.1088
1	(17)	1.0084	0.0000e+00	1.4292	0.2226
1	(18)	1.0084	0.0000e+00	1.4292	0.2226
1	(19)	1.0084	0.0000e+00	1.4292	0.2226
2	(14)	1.0975	-5.1323e-02	0.8810	-0.0235
2	(15)	1.0750	-3.1951e-02	0.9000	-0.0260
2	(17)	1.0049	-3.6885e-02	0.9000	0.0399
2	(18)	1.0049	-3.6885e-02	0.9000	0.0399
2	(19)	1.0049	-3.6885e-02	0.9000	0.0399
3	(14)	1.0876	-5.5535e-02	0.7157	-0.2181
3	(15)	1.0750	-2.5267e-02	0.8500	-0.1374

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
3	(17)	1.0070	-2.9540e-02	0.8500	-0.0524
3	(18)	1.0070	-2.9540e-02	0.8500	-0.0524
3	(19)	1.0070	-2.9540e-02	0.8500	-0.0524
4	(14)	1.0968	-7.5339e-02	—	—
4	(15)	1.0719	-7.1288e-02	—	—
4	(17)	0.9991	-8.1798e-02	—	—
4	(18)	0.9991	-8.1798e-02	—	—
4	(19)	0.9991	-8.1798e-02	—	—
5	(14)	1.0756	-1.4017e-01	—	—
5	(15)	1.0501	-1.3446e-01	—	—
5	(17)	0.9725	-1.5478e-01	—	—
5	(18)	0.9725	-1.5478e-01	—	—
5	(19)	0.9725	-1.5478e-01	—	—
6	(14)	1.0873	-1.2573e-01	—	—
6	(15)	1.0642	-1.1692e-01	—	—
6	(17)	0.9880	-1.3418e-01	—	—
6	(18)	0.9880	-1.3418e-01	—	—
6	(19)	0.9880	-1.3418e-01	—	—
7	(14)	1.1000	-9.6946e-02	—	—
7	(15)	1.0778	-8.0519e-02	—	—
7	(17)	1.0039	-9.2674e-02	—	—
7	(18)	1.0039	-9.2674e-02	—	—
7	(19)	1.0039	-9.2674e-02	—	—
8	(14)	1.0895	-1.2879e-01	—	—
8	(15)	1.0691	-1.1106e-01	—	—
8	(17)	0.9938	-1.2786e-01	—	—
8	(18)	0.9938	-1.2786e-01	—	—
8	(19)	0.9938	-1.2786e-01	—	—
9	(14)	1.1000	-9.0598e-02	—	—
9	(15)	1.0835	-6.8045e-02	—	—
9	(17)	1.0112	-7.8477e-02	—	—
9	(18)	1.0112	-7.8477e-02	—	—
9	(19)	1.0112	-7.8477e-02	—	—

Caso 9bus_transformer_fields.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
1	(14)	1.1000	-1.0160e-30	1.5684	0.0825
1	(15)	1.0750	0.0000e+00	1.4249	0.1088
1	(17)	1.0352	0.0000e+00	1.4278	0.2618
1	(18)	1.0106	0.0000e+00	1.4273	0.3571
1	(19)	1.0106	0.0000e+00	1.4273	0.3571
2	(14)	1.1000	-5.0960e-02	0.8795	-0.0472
2	(15)	1.0750	-3.1951e-02	0.9000	-0.0260
2	(17)	1.0164	-3.1430e-02	0.9000	-0.0217
2	(18)	1.0039	-2.5650e-02	0.9000	-0.0524
2	(19)	1.0039	-2.5650e-02	0.9000	-0.0524
3	(14)	1.0982	-5.5124e-02	0.7252	-0.1733
3	(15)	1.0750	-2.5267e-02	0.8500	-0.1374
3	(17)	1.0204	-2.4604e-02	0.8500	-0.0985
3	(18)	1.0177	-1.7382e-02	0.8500	-0.1963
3	(19)	1.0177	-1.7382e-02	0.8500	-0.1963

Barra	Script	vm (pu)	va (rad)	pg (pu)	qg (pu)
4	(14)	1.0988	-7.4818e-02	—	—
4	(15)	1.0719	-7.1288e-02	—	—
4	(17)	1.0237	-7.7681e-02	—	—
4	(18)	1.0433	-7.4428e-02	—	—
4	(19)	1.0433	-7.4428e-02	—	—
5	(14)	1.0783	-1.3930e-01	—	—
5	(15)	1.0501	-1.3446e-01	—	—
5	(17)	0.9953	-1.4693e-01	—	—
5	(18)	1.0117	-1.4077e-01	—	—
5	(19)	1.0117	-1.4077e-01	—	—
6	(14)	1.0916	-1.2484e-01	—	—
6	(15)	1.0642	-1.1692e-01	—	—
6	(17)	1.0107	-1.2720e-01	—	—
6	(18)	1.0251	-1.2144e-01	—	—
6	(19)	1.0251	-1.2144e-01	—	—
7	(14)	1.1038	-9.6245e-02	—	—
7	(15)	1.0778	-8.0519e-02	—	—
7	(17)	1.0192	-8.5757e-02	—	—
7	(18)	1.0270	-7.9254e-02	—	—
7	(19)	1.0270	-7.9254e-02	—	—
8	(14)	1.0953	-1.2779e-01	—	—
8	(15)	1.0691	-1.1106e-01	—	—
8	(17)	1.0098	-1.1994e-01	—	—
8	(18)	1.0158	-1.1270e-01	—	—
8	(19)	1.0158	-1.1270e-01	—	—
9	(14)	1.1081	-9.0051e-02	—	—
9	(15)	1.0835	-6.8045e-02	—	—
9	(17)	1.0272	-7.2141e-02	—	—
9	(18)	1.0302	-6.4908e-02	—	—
9	(19)	1.0302	-6.4908e-02	—	—

Caso test_defaults.pwf

Nenhuma referência (14) ou (15) convergiu para este caso; análise pulada.

Caso test_line_shunt.pwf

Nenhuma referência (14) ou (15) convergiu para este caso; análise pulada.

Caso test_system.pwf

Nenhuma referência (14) ou (15) convergiu para este caso; análise pulada.

3. Fluxos nos Ramos

Mesmo critério da Seção 2 aplicado a **pf_pu**, **qf_pu**, **pt_pu**, **qt_pu** (excede 0.0001 em AMBAS (14) E (15)). Cada ramo divergente é exibido com uma linha para (14) e (15) e uma para cada script (16)–(19) que divergiu. Scripts (16)–(19) numericamente equivalentes (mesmo cluster) a (14) ou (15) são omitidos do caso. A coluna **Ramo (f→t)** traz o **branch_id** seguido das barras de envio (**f_bus**) e recepção (**t_bus**); **pf/qf** são os fluxos no terminal **f** e **pt/qt** no terminal **t**.

Caso 01 MAXIMA NOTURNA_DEZ25.PWF

Sem referência convergida; análise pulada.

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
1 (206→207)	(15)	-0.0880	-0.0898	0.0891	0.0913
1 (206→207)	(16)	-0.0967	-0.1066	0.0980	0.1082
1 (206→207)	(17)	-0.0906	-0.1012	0.0918	0.1027
1 (206→207)	(18)	-0.1521	-0.1480	0.1550	0.1516
1 (206→207)	(19)	-0.1521	-0.1495	0.1550	0.1532
2 (9021→9023)	(15)	0.0091	0.0035	-0.0090	-0.0034
2 (9021→9023)	(16)	0.0096	-0.0216	-0.0090	0.0220
2 (9021→9023)	(17)	0.0093	-0.0163	-0.0090	0.0166
2 (9021→9023)	(18)	0.0091	0.0026	-0.0090	-0.0026
2 (9021→9023)	(19)	0.0091	0.0027	-0.0090	-0.0026
3 (1→5)	(15)	4.0627	1.0394	-4.0459	-0.9387
3 (1→5)	(16)	4.0700	1.1219	-4.0545	-1.0285
3 (1→5)	(17)	4.0671	1.0531	-4.0518	-0.9611
3 (1→5)	(18)	4.0821	1.3293	-4.0659	-1.2319
3 (1→5)	(19)	4.0820	1.3413	-4.0658	-1.2438
4 (37→38)	(15)	-0.3084	0.0132	0.3092	-0.0165
4 (37→38)	(16)	-0.3065	-0.1929	0.3076	0.1908
4 (37→38)	(17)	-0.3055	-0.2073	0.3066	0.2054
4 (37→38)	(18)	-0.2561	-0.3421	0.2575	0.3422
4 (37→38)	(19)	-0.2591	-0.3440	0.2606	0.3443
5 (70→528)	(15)	0.2837	-0.2141	-0.2810	0.2050
5 (70→528)	(16)	0.2835	-0.2344	-0.2810	0.2196
5 (70→528)	(17)	0.2835	-0.2385	-0.2810	0.2232
5 (70→528)	(18)	0.2836	-0.2157	-0.2810	0.2051
5 (70→528)	(19)	0.2836	-0.2151	-0.2810	0.2051
6 (25→26)	(15)	0.7875	-0.0344	-0.7661	0.0418
6 (25→26)	(16)	0.7885	-0.0360	-0.7691	0.0359
6 (25→26)	(17)	0.7866	-0.0335	-0.7673	0.0332
6 (25→26)	(18)	0.8323	0.1540	-0.8083	-0.1422
6 (25→26)	(19)	0.8322	0.1535	-0.8082	-0.1418
7 (41→42)	(15)	-5.5989	0.3077	5.5989	0.5998
7 (41→42)	(16)	-5.5206	-0.3408	5.5206	1.1942
7 (41→42)	(17)	-5.5116	-0.4169	5.5116	1.2653
7 (41→42)	(18)	-5.3092	-0.9044	5.3092	1.7000
7 (41→42)	(19)	-5.3136	-0.9161	5.3136	1.7133
8 (231→232)	(15)	5.5289	0.9893	-5.5228	-2.2713
8 (231→232)	(16)	5.5314	1.6176	-5.5253	-3.0090
8 (231→232)	(17)	5.5267	-0.1917	-5.5216	-1.2688
8 (231→232)	(18)	5.5281	-0.7604	-5.5227	-0.5932
8 (231→232)	(19)	5.5280	-0.7288	-5.5226	-0.6242
9 (201→204)	(15)	-0.1193	0.0639	0.1217	-0.0603
9 (201→204)	(16)	-0.1485	0.0499	0.1513	-0.0457
9 (201→204)	(17)	-0.1525	0.0572	0.1554	-0.0528
9 (201→204)	(18)	-0.2120	-0.2535	0.2273	0.2761
9 (201→204)	(19)	-0.2135	-0.2580	0.2293	0.2813
10 (119→1190)	(15)	1.0040	0.3151	-1.0030	-0.2917
10 (119→1190)	(16)	1.0039	0.3131	-1.0030	-0.2918
10 (119→1190)	(17)	1.0039	0.3133	-1.0030	-0.2918
10 (119→1190)	(18)	1.0040	0.3143	-1.0030	-0.2917
10 (119→1190)	(19)	1.0040	0.3143	-1.0030	-0.2917
11 (115→122)	(15)	1.8322	0.7311	-1.8156	-0.6429
11 (115→122)	(16)	1.9268	1.0257	-1.9085	-0.9284
11 (115→122)	(17)	1.9108	0.9454	-1.8934	-0.8532

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
11 (115→122)	(18)	1.7363	0.6480	-1.7217	-0.5705
11 (115→122)	(19)	1.7364	0.6495	-1.7218	-0.5720
12 (9006→9007)	(15)	0.0913	0.0280	-0.0908	-0.0256
12 (9006→9007)	(16)	0.0912	-0.0607	-0.0906	0.0632
12 (9006→9007)	(17)	0.0912	-0.0594	-0.0906	0.0618
12 (9006→9007)	(18)	0.0911	0.0224	-0.0906	-0.0203
12 (9006→9007)	(19)	0.0911	0.0225	-0.0906	-0.0205
13 (145→146)	(15)	-2.0513	-0.4036	2.0876	0.5627
13 (145→146)	(16)	-2.0738	-0.0775	2.1089	0.2270
13 (145→146)	(17)	-2.0647	-0.3370	2.1007	0.4936
13 (145→146)	(18)	-2.0585	-0.7294	2.0968	0.8979
13 (145→146)	(19)	-2.0863	-1.2412	2.1399	1.5228
14 (228→231)	(15)	0.0546	-0.1151	-0.0546	0.1165
14 (228→231)	(16)	0.0573	-0.3076	-0.0570	0.3155
14 (228→231)	(17)	0.0338	-0.8480	-0.0320	0.9113
14 (228→231)	(18)	0.0226	-0.8784	-0.0206	0.9520
14 (228→231)	(19)	0.0226	-0.8783	-0.0205	0.9519
15 (129→133)	(15)	0.9898	-0.1564	-0.9877	0.1457
15 (129→133)	(16)	1.0094	0.1313	-1.0072	-0.1435
15 (129→133)	(17)	1.0277	-0.1181	-1.0256	0.1041
15 (129→133)	(18)	1.0343	-0.0793	-1.0322	0.0647
15 (129→133)	(19)	1.0435	-0.0707	-1.0414	0.0565
16 (189→208)	(15)	0.0802	0.0248	-0.0800	-0.0242
16 (189→208)	(16)	0.0626	0.0415	-0.0624	-0.0411
16 (189→208)	(17)	0.0623	0.0435	-0.0621	-0.0430
16 (189→208)	(18)	0.0359	-0.0623	-0.0357	0.0627
16 (189→208)	(19)	0.0369	-0.0616	-0.0368	0.0620
17 (57→58)	(15)	-0.5951	0.8085	0.6406	-0.7658
17 (57→58)	(16)	-0.6465	0.6237	0.6792	-0.6084
17 (57→58)	(17)	-0.6384	0.6247	0.6705	-0.6106
17 (57→58)	(18)	-0.6060	0.6650	0.6384	-0.6506
17 (57→58)	(19)	-0.6117	0.6656	0.6444	-0.6506
18 (200→210)	(15)	-0.2345	-0.0311	0.2394	0.0258
18 (200→210)	(16)	-0.2513	-0.1300	0.2574	0.1245
18 (200→210)	(17)	-0.2350	-0.0352	0.2392	0.0262
18 (200→210)	(18)	-0.2000	0.1111	0.2040	-0.1200
18 (200→210)	(19)	-0.1999	0.1108	0.2039	-0.1197
19 (15→89)	(15)	0.4519	-0.1944	-0.4288	0.2166
19 (15→89)	(16)	0.4404	-0.2376	-0.4180	0.2560
19 (15→89)	(17)	0.4381	-0.2267	-0.4164	0.2429
19 (15→89)	(18)	0.4949	-0.0812	-0.4741	0.0928
19 (15→89)	(19)	0.4904	-0.0874	-0.4698	0.0985
20 (197→198)	(15)	-0.9520	-0.2430	0.9618	0.2930
20 (197→198)	(16)	-0.9566	-0.4562	0.9665	0.5049
20 (197→198)	(17)	-0.9617	-0.3372	0.9707	0.3807
20 (197→198)	(18)	-0.9506	-0.7767	0.9643	0.8486
20 (197→198)	(19)	-0.9505	-0.7771	0.9642	0.8490

(mostrando 20 de 411 ramos divergentes; tabela truncada)

Caso 3bus.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
1 (1→2)	(14)	0.0000	0.0000	-0.0000	-0.0000
1 (1→2)	(15)	0.0179	-0.0243	-0.0177	0.0245
1 (1→2)	(17)	0.0179	-0.0202	-0.0178	0.0204
1 (1→2)	(18)	0.0179	-0.0202	-0.0178	0.0204
1 (1→2)	(19)	0.0179	-0.0202	-0.0178	0.0204
2 (1→3)	(14)	0.1557	0.0303	-0.1520	-0.0266
2 (1→3)	(15)	0.1650	0.0192	-0.1603	-0.0146
2 (1→3)	(17)	0.1653	0.0214	-0.1604	-0.0166
2 (1→3)	(18)	0.1653	0.0214	-0.1604	-0.0166
2 (1→3)	(19)	0.1653	0.0214	-0.1604	-0.0166
3 (2→3)	(14)	0.1557	0.0303	-0.1519	-0.0266
3 (2→3)	(15)	0.1476	0.0425	-0.1436	-0.0386
3 (2→3)	(17)	0.1477	0.0407	-0.1435	-0.0367
3 (2→3)	(18)	0.1477	0.0407	-0.1435	-0.0367
3 (2→3)	(19)	0.1477	0.0407	-0.1435	-0.0367

Caso 3bus_DBSH.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
1 (1→2)	(14)	-0.0000	-0.0000	0.0000	0.0000
1 (1→2)	(15)	0.0254	-0.0320	-0.0251	0.0323
1 (1→2)	(17)	0.0255	-0.0264	-0.0253	0.0265
1 (1→2)	(18)	0.0255	-0.0264	-0.0253	0.0265
1 (1→2)	(19)	0.0255	-0.0263	-0.0253	0.0265
2 (1→3)	(14)	0.1660	0.2135	-0.1519	-0.2000
2 (1→3)	(15)	0.1778	0.2224	-0.1639	-0.2091
2 (1→3)	(17)	0.1786	0.2555	-0.1641	-0.2415
2 (1→3)	(18)	0.1786	0.2555	-0.1641	-0.2415
2 (1→3)	(19)	0.1786	0.2555	-0.1641	-0.2415
3 (2→3)	(14)	0.1660	0.2135	-0.1519	-0.2000
3 (2→3)	(15)	0.1550	0.2532	-0.1400	-0.2388
3 (2→3)	(17)	0.1552	0.2807	-0.1398	-0.2659
3 (2→3)	(18)	0.1552	0.2807	-0.1398	-0.2659
3 (2→3)	(19)	0.1552	0.2806	-0.1398	-0.2658

Caso 3bus_DCER.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
1 (1→2)	(14)	-0.0000	-0.0000	0.0000	0.0000
1 (1→2)	(15)	0.0254	-0.0320	-0.0251	0.0323
1 (1→2)	(17)	0.0271	-0.0738	-0.0261	0.0749
1 (1→2)	(18)	0.0271	-0.0738	-0.0261	0.0749
1 (1→2)	(19)	0.0271	-0.0738	-0.0261	0.0749
2 (1→3)	(14)	0.1660	0.2135	-0.1519	-0.2000
2 (1→3)	(15)	0.1778	0.2224	-0.1639	-0.2091
2 (1→3)	(17)	0.1772	0.1999	-0.1648	-0.1880
2 (1→3)	(18)	0.1772	0.1999	-0.1648	-0.1880
2 (1→3)	(19)	0.1772	0.1999	-0.1648	-0.1880

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
3 (2→3)	(14)	0.1660	0.2135	-0.1519	-0.2000
3 (2→3)	(15)	0.1550	0.2532	-0.1400	-0.2388
3 (2→3)	(17)	0.1560	0.2732	-0.1391	-0.2570
3 (2→3)	(18)	0.1560	0.2732	-0.1391	-0.2570
3 (2→3)	(19)	0.1560	0.2732	-0.1391	-0.2570

Caso 3bus_DCSC.pwf

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
1 (1→3)	(14)	0.0003	0.0142	-0.0002	-0.0141
1 (1→3)	(15)	0.0887	-0.0953	-0.0858	0.0981
1 (1→3)	(16)	0.0872	0.0131	-0.0859	-0.0119
1 (1→3)	(17)	0.0890	0.0153	-0.0875	-0.0139
1 (1→3)	(18)	0.0890	0.0153	-0.0875	-0.0139
1 (1→3)	(19)	0.0890	0.0153	-0.0875	-0.0139
2 (2→3)	(14)	0.0003	0.0142	-0.0002	-0.0141
2 (2→3)	(15)	0.0910	-0.0918	-0.0882	0.0946
2 (2→3)	(16)	0.0895	0.0167	-0.0881	-0.0154
2 (2→3)	(17)	0.0879	0.0157	-0.0865	-0.0143
2 (2→3)	(18)	0.0879	0.0157	-0.0865	-0.0143
2 (2→3)	(19)	0.0879	0.0157	-0.0865	-0.0143
3 (1→2)	(14)	-0.0000	-0.0000	0.0000	0.0000
3 (1→2)	(15)	-0.0389	0.1615	0.0389	-0.1617
3 (1→2)	(16)	-0.0404	0.1625	0.0404	-0.1627
3 (1→2)	(17)	-0.0420	-0.0208	0.0420	0.0208
3 (1→2)	(18)	-0.0420	-0.0208	0.0420	0.0208
3 (1→2)	(19)	-0.0420	-0.0208	0.0420	0.0208

Caso 3bus_DCline.pwf

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
1 (1→3)	(14)	0.1557	0.0303	-0.1519	-0.0266
1 (1→3)	(16)	0.1721	1.1852	0.0425	-0.9789
1 (1→3)	(17)	0.1721	1.1852	0.0425	-0.9789
1 (1→3)	(18)	0.1721	1.1852	0.0425	-0.9789
1 (1→3)	(19)	0.1721	1.1852	0.0425	-0.9789
2 (2→3)	(14)	0.1557	0.0303	-0.1520	-0.0266
2 (2→3)	(16)	6.2663	0.5556	-0.3464	5.1354
2 (2→3)	(17)	6.2663	0.5556	-0.3464	5.1354
2 (2→3)	(18)	6.2663	0.5556	-0.3464	5.1354
2 (2→3)	(19)	6.2663	0.5556	-0.3464	5.1354

Caso 3bus_DSHL.pwf

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
1 (1→2)	(15)	0.5778	1.7819	0.0220	-1.2053
1 (1→2)	(16)	0.1160	0.8040	-0.0068	-0.6990
1 (1→2)	(17)	0.1160	0.8041	-0.0068	-0.6991
1 (1→2)	(18)	0.1160	0.8041	-0.0068	-0.6991
1 (1→2)	(19)	0.1160	0.8041	-0.0068	-0.6991

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
2 (1→3)	(15)	0.6266	1.4089	-0.2202	-1.0182
2 (1→3)	(16)	0.2354	0.5000	-0.1849	-0.4514
2 (1→3)	(17)	0.2354	0.4999	-0.1849	-0.4513
2 (1→3)	(18)	0.2354	0.4999	-0.1849	-0.4513
2 (1→3)	(19)	0.2354	0.4999	-0.1849	-0.4513
3 (2→3)	(15)	0.1079	1.2303	-0.0837	0.9649
3 (2→3)	(16)	0.1367	2.4275	-0.1190	1.5407
3 (2→3)	(17)	0.1367	2.4272	-0.1190	1.5411
3 (2→3)	(18)	0.1367	2.4272	-0.1190	1.5411
3 (2→3)	(19)	0.1367	2.4272	-0.1190	1.5411

Caso 3bus_corrections.pwf

Sem referência convergida; análise pulada.

Caso 3bus_shunt_fields.pwf

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
1 (1→2)	(14)	0.0163	0.2105	-0.0072	-0.2018
1 (1→2)	(15)	0.0181	-0.0246	-0.0180	0.0248
1 (1→2)	(16)	0.0274	0.2543	-0.0167	-0.2440
1 (1→2)	(17)	0.0238	0.1643	-0.0192	-0.1599
1 (1→2)	(18)	0.0238	0.1643	-0.0192	-0.1599
1 (1→2)	(19)	0.0238	0.1643	-0.0192	-0.1599
2 (1→3)	(14)	0.1684	0.0521	-0.1621	-0.0460
2 (1→3)	(15)	0.1663	-0.0831	-0.1604	0.0888
2 (1→3)	(16)	0.1729	0.0505	-0.1676	-0.0454
2 (1→3)	(17)	0.1721	-0.0558	-0.1666	0.0611
2 (1→3)	(18)	0.1721	-0.0558	-0.1666	0.0611
2 (1→3)	(19)	0.1721	-0.0558	-0.1666	0.0611
3 (2→3)	(14)	0.1517	-0.1453	-0.1418	0.1547
3 (2→3)	(15)	0.1479	-0.0596	-0.1435	0.0638
3 (2→3)	(16)	0.1466	-0.1893	-0.1363	0.1992
3 (2→3)	(17)	0.1491	-0.2099	-0.1373	0.2213
3 (2→3)	(18)	0.1491	-0.2099	-0.1373	0.2213
3 (2→3)	(19)	0.1491	-0.2099	-0.1373	0.2213

Caso 3busfrank.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
1 (1→2)	(14)	0.0000	-0.0000	-0.0000	0.0000
1 (1→2)	(15)	0.0726	-0.0794	-0.0706	0.0813
1 (1→2)	(17)	0.0711	-0.0380	-0.0700	0.0391
1 (1→2)	(18)	0.0711	-0.0380	-0.0700	0.0391
1 (1→2)	(19)	0.0711	-0.0380	-0.0700	0.0391
2 (1→3)	(14)	0.2324	0.1241	-0.2220	-0.1141
2 (1→3)	(15)	0.2686	0.0889	-0.2549	-0.0757
2 (1→3)	(17)	0.2695	0.1096	-0.2548	-0.0954
2 (1→3)	(18)	0.2695	0.1096	-0.2548	-0.0954
2 (1→3)	(19)	0.2695	0.1096	-0.2548	-0.0954

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
3 (2→3)	(14)	0.2324	0.1241	-0.2220	-0.1141
3 (2→3)	(15)	0.2005	0.1634	-0.1891	-0.1525
3 (2→3)	(17)	0.1999	0.1431	-0.1892	-0.1328
3 (2→3)	(18)	0.1999	0.1431	-0.1892	-0.1328
3 (2→3)	(19)	0.1999	0.1431	-0.1892	-0.1328

Caso 3busfrank_continuous_shunt.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
1 (1→2)	(14)	-0.0000	0.0000	0.0000	-0.0000
1 (1→2)	(15)	0.0346	-0.0414	-0.0341	0.0419
1 (1→2)	(17)	0.0339	-0.0078	-0.0337	0.0080
1 (1→2)	(18)	0.0339	-0.0078	-0.0337	0.0080
1 (1→2)	(19)	0.0339	-0.0078	-0.0337	0.0080
2 (1→2)	(14)	-0.0000	0.0000	0.0000	-0.0000
2 (1→2)	(15)	0.0346	-0.0414	-0.0341	0.0419
2 (1→2)	(17)	0.0339	-0.0078	-0.0337	0.0080
2 (1→2)	(18)	0.0339	-0.0078	-0.0337	0.0080
2 (1→2)	(19)	0.0339	-0.0078	-0.0337	0.0080
3 (1→3)	(14)	0.1150	0.1174	-0.1110	-0.1135
3 (1→3)	(15)	0.1319	0.0903	-0.1275	-0.0861
3 (1→3)	(17)	0.1322	0.1063	-0.1273	-0.1016
3 (1→3)	(18)	0.1322	0.1063	-0.1273	-0.1016
3 (1→3)	(19)	0.1322	0.1063	-0.1273	-0.1016
4 (1→3)	(14)	0.1150	0.1174	-0.1110	-0.1135
4 (1→3)	(15)	0.1319	0.0903	-0.1275	-0.0861
4 (1→3)	(17)	0.1322	0.1063	-0.1273	-0.1016
4 (1→3)	(18)	0.1322	0.1063	-0.1273	-0.1016
4 (1→3)	(19)	0.1322	0.1063	-0.1273	-0.1016
5 (2→3)	(14)	0.1150	0.1174	-0.1110	-0.1135
5 (2→3)	(15)	0.0991	0.1306	-0.0945	-0.1262
5 (2→3)	(17)	0.0987	0.1130	-0.0947	-0.1092
5 (2→3)	(18)	0.0987	0.1130	-0.0947	-0.1092
5 (2→3)	(19)	0.0987	0.1130	-0.0947	-0.1092
6 (2→3)	(14)	0.1150	0.1174	-0.1110	-0.1135
6 (2→3)	(15)	0.0991	0.1306	-0.0945	-0.1262
6 (2→3)	(17)	0.0987	0.1130	-0.0947	-0.1092
6 (2→3)	(18)	0.0987	0.1130	-0.0947	-0.1092
6 (2→3)	(19)	0.0987	0.1130	-0.0947	-0.1092

Caso 3busfrank_qlim.pwf

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
1 (1→2)	(14)	-0.0000	0.0000	0.0000	-0.0000
1 (1→2)	(15)	0.1484	-0.1525	-0.1406	0.1599
1 (1→2)	(16)	0.1388	-0.0926	-0.1342	0.0970
1 (1→2)	(17)	0.1290	0.0584	-0.1259	-0.0554
1 (1→2)	(18)	0.1290	0.0584	-0.1259	-0.0554
1 (1→2)	(19)	0.1290	0.0584	-0.1259	-0.0554
2 (1→3)	(14)	0.3137	0.5609	-0.2519	-0.5015

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
2 (1→3)	(15)	0.3858	0.5046	-0.3168	-0.4383
2 (1→3)	(16)	0.3819	0.5284	-0.3125	-0.4617
2 (1→3)	(17)	0.3866	0.6065	-0.3056	-0.5286
2 (1→3)	(18)	0.3866	0.6065	-0.3056	-0.5286
2 (1→3)	(19)	0.3866	0.6065	-0.3056	-0.5286
3 (2→3)	(14)	0.3137	0.5609	-0.2520	-0.5015
3 (2→3)	(15)	0.2705	0.6449	-0.1871	-0.5647
3 (2→3)	(16)	0.2641	0.6070	-0.1914	-0.5372
3 (2→3)	(17)	0.2558	0.5296	-0.1983	-0.4744
3 (2→3)	(18)	0.2558	0.5296	-0.1983	-0.4744
3 (2→3)	(19)	0.2558	0.5296	-0.1983	-0.4744

Caso 4busfrank_vlim.pwf

Sem referência convergida; análise pulada.

Caso 500bus.pwf

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
1 (242→287)	(15)	-0.8759	-0.0131	0.8813	0.0430
1 (242→287)	(16)	-0.8760	-0.0131	0.8814	0.0429
1 (242→287)	(17)	-0.8828	-0.0615	0.8885	0.0941
1 (242→287)	(18)	-0.8828	-0.0615	0.8885	0.0941
1 (242→287)	(19)	-0.8828	-0.0615	0.8885	0.0941
2 (325→324)	(15)	-1.0350	-0.2759	1.0356	0.2802
2 (325→324)	(16)	-1.0350	-0.2761	1.0356	0.2805
2 (325→324)	(17)	-1.0350	-0.2759	1.0357	0.2804
2 (325→324)	(18)	-1.0350	-0.2759	1.0357	0.2804
2 (325→324)	(19)	-1.0350	-0.2759	1.0357	0.2804
5 (423→496)	(15)	-1.8987	-0.1652	1.9124	0.2846
5 (423→496)	(16)	-1.8988	-0.1657	1.9125	0.2852
5 (423→496)	(17)	-1.9015	-0.2173	1.9156	0.3411
5 (423→496)	(18)	-1.9015	-0.2173	1.9156	0.3411
5 (423→496)	(19)	-1.9015	-0.2173	1.9156	0.3411
7 (294→293)	(15)	-0.2561	-0.0683	0.2561	0.0685
7 (294→293)	(16)	-0.2561	-0.0685	0.2561	0.0687
8 (27→382)	(15)	0.4221	0.3282	-0.4212	-0.3315
8 (27→382)	(16)	0.4230	0.3333	-0.4221	-0.3366
8 (27→382)	(17)	0.4102	0.2572	-0.4095	-0.2610
8 (27→382)	(18)	0.4102	0.2572	-0.4095	-0.2610
8 (27→382)	(19)	0.4102	0.2572	-0.4095	-0.2610
10 (46→156)	(15)	0.3893	0.1363	-0.3882	-0.1455
10 (46→156)	(16)	0.3893	0.1361	-0.3881	-0.1452
10 (46→156)	(17)	0.3880	0.1320	-0.3868	-0.1398
10 (46→156)	(18)	0.3880	0.1320	-0.3868	-0.1398
10 (46→156)	(19)	0.3880	0.1320	-0.3868	-0.1398
11 (387→470)	(15)	0.9082	0.1488	-0.9065	-0.1401
11 (387→470)	(16)	0.9083	0.1504	-0.9065	-0.1416
11 (387→470)	(17)	0.9080	0.1553	-0.9062	-0.1456
11 (387→470)	(18)	0.9080	0.1553	-0.9062	-0.1456
11 (387→470)	(19)	0.9080	0.1553	-0.9062	-0.1456
12 (495→449)	(15)	-0.4031	-0.1106	0.4034	0.1075
12 (495→449)	(16)	-0.4029	-0.1100	0.4032	0.1069

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
12 (495→449)	(17)	-0.4012	-0.0991	0.4015	0.0963
12 (495→449)	(18)	-0.4012	-0.0991	0.4015	0.0963
12 (495→449)	(19)	-0.4012	-0.0991	0.4015	0.0963
13 (365→374)	(15)	0.3016	0.0814	-0.3013	-0.0875
13 (365→374)	(16)	0.3015	0.0809	-0.3012	-0.0871
13 (365→374)	(17)	0.2966	0.0441	-0.2964	-0.0501
13 (365→374)	(18)	0.2966	0.0441	-0.2964	-0.0501
13 (365→374)	(19)	0.2966	0.0441	-0.2964	-0.0501
14 (263→491)	(15)	0.9874	0.0720	-0.9841	-0.0515
14 (263→491)	(16)	0.9875	0.0717	-0.9842	-0.0512
14 (263→491)	(17)	0.9904	0.1041	-0.9869	-0.0819
14 (263→491)	(18)	0.9904	0.1041	-0.9869	-0.0819
14 (263→491)	(19)	0.9904	0.1041	-0.9869	-0.0819
15 (285→487)	(15)	1.0728	-0.2955	-1.0691	0.3211
15 (285→487)	(16)	1.0743	-0.2911	-1.0705	0.3165
15 (285→487)	(17)	1.0746	-0.2073	-1.0708	0.2342
15 (285→487)	(18)	1.0746	-0.2073	-1.0708	0.2342
15 (285→487)	(19)	1.0746	-0.2073	-1.0708	0.2342
16 (401→5)	(15)	1.2048	-0.2458	-1.1976	0.2999
16 (401→5)	(16)	1.2047	-0.2438	-1.1975	0.2979
16 (401→5)	(17)	1.2058	-0.1532	-1.1985	0.2080
16 (401→5)	(18)	1.2058	-0.1532	-1.1985	0.2080
16 (401→5)	(19)	1.2058	-0.1532	-1.1985	0.2080
18 (133→369)	(15)	-0.3184	-0.0592	0.3190	0.0442
18 (133→369)	(16)	-0.3185	-0.0609	0.3191	0.0460
18 (133→369)	(17)	-0.3190	-0.0754	0.3196	0.0615
18 (133→369)	(18)	-0.3190	-0.0754	0.3196	0.0615
18 (133→369)	(19)	-0.3190	-0.0754	0.3196	0.0615
19 (493→380)	(15)	0.8658	0.0888	-0.8633	-0.0755
19 (493→380)	(16)	0.8664	0.0967	-0.8639	-0.0835
19 (493→380)	(17)	0.8599	0.0905	-0.8573	-0.0756
19 (493→380)	(18)	0.8599	0.0905	-0.8573	-0.0756
19 (493→380)	(19)	0.8599	0.0905	-0.8573	-0.0756
20 (322→208)	(15)	1.2613	-0.6731	-1.2555	0.7198
20 (322→208)	(16)	1.2605	-0.6757	-1.2546	0.7225
20 (322→208)	(17)	1.2729	-0.5466	-1.2672	0.5926
20 (322→208)	(18)	1.2729	-0.5466	-1.2672	0.5926
20 (322→208)	(19)	1.2729	-0.5466	-1.2672	0.5926
22 (263→276)	(15)	1.1648	0.6242	-1.1633	-0.6129
22 (263→276)	(16)	1.1652	0.6258	-1.1637	-0.6145
22 (263→276)	(17)	1.1682	0.6593	-1.1666	-0.6471
22 (263→276)	(18)	1.1682	0.6593	-1.1666	-0.6471
22 (263→276)	(19)	1.1682	0.6593	-1.1666	-0.6471
25 (217→216)	(15)	-0.5510	-0.1469	0.5512	0.1479
25 (217→216)	(16)	-0.5510	-0.1472	0.5512	0.1482
26 (173→172)	(15)	-0.0314	-0.0084	0.0314	0.0084
26 (173→172)	(16)	-0.0314	-0.0085	0.0314	0.0085
27 (368→367)	(15)	-0.1145	-0.0305	0.1145	0.0306
27 (368→367)	(16)	-0.1145	-0.0307	0.1145	0.0308
28 (66→308)	(15)	0.5667	0.0384	-0.5644	-0.0404
28 (66→308)	(16)	0.5671	0.0257	-0.5649	-0.0278
28 (66→308)	(17)	0.5632	0.0259	-0.5608	-0.0250
28 (66→308)	(18)	0.5632	0.0259	-0.5608	-0.0250
28 (66→308)	(19)	0.5632	0.0259	-0.5608	-0.0250

(mostrando 20 de 509 ramos divergentes; tabela truncada)

Caso 5busfrank.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
2 (1→2)	(14)	0.0000	-0.0000	-0.0000	0.0000
2 (1→2)	(15)	0.0770	-0.0838	-0.0748	0.0859
2 (1→2)	(17)	0.0750	-0.0436	-0.0738	0.0448
2 (1→2)	(18)	0.0750	-0.0436	-0.0738	0.0448
2 (1→2)	(19)	0.0750	-0.0436	-0.0738	0.0448
4 (1→3)	(14)	0.2379	0.1489	-0.2261	-0.1375
4 (1→3)	(15)	0.2762	0.1120	-0.2610	-0.0974
4 (1→3)	(17)	0.2764	0.1315	-0.2607	-0.1164
4 (1→3)	(18)	0.2764	0.1315	-0.2607	-0.1164
4 (1→3)	(19)	0.2764	0.1315	-0.2607	-0.1164
5 (2→3)	(14)	0.2379	0.1489	-0.2261	-0.1375
5 (2→3)	(15)	0.2047	0.1905	-0.1913	-0.1777
5 (2→3)	(17)	0.2037	0.1702	-0.1917	-0.1587
5 (2→3)	(18)	0.2037	0.1702	-0.1917	-0.1587
5 (2→3)	(19)	0.2037	0.1702	-0.1917	-0.1587

Caso 5busfrank_cphs.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
2 (1→2)	(14)	-0.0000	-0.0000	0.0000	0.0000
2 (1→2)	(15)	0.1168	-0.1225	-0.1119	0.1272
2 (1→2)	(17)	0.1216	-0.1853	-0.1132	0.1934
2 (1→2)	(18)	0.1216	-0.1853	-0.1132	0.1934
2 (1→2)	(19)	0.1216	-0.1853	-0.1132	0.1934
4 (1→3)	(14)	0.2923	0.1203	-0.2774	-0.1060
4 (1→3)	(15)	0.3509	0.0676	-0.3291	-0.0467
4 (1→3)	(17)	0.3509	0.0370	-0.3297	-0.0166
4 (1→3)	(18)	0.3509	0.0370	-0.3297	-0.0166
4 (1→3)	(19)	0.3509	0.0370	-0.3297	-0.0166
5 (2→3)	(14)	0.2923	0.1203	-0.2774	-0.1060
5 (2→3)	(15)	0.2418	0.1808	-0.2262	-0.1658
5 (2→3)	(17)	0.2431	0.2126	-0.2256	-0.1959
5 (2→3)	(18)	0.2431	0.2126	-0.2256	-0.1959
5 (2→3)	(19)	0.2431	0.2126	-0.2256	-0.1959

Caso 5busfrank_csca.pwf

Sem referência convergida; análise pulada.

Caso 5busfrank_ctaf.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
1 (1→2)	(14)	-0.0000	-0.0000	0.0000	0.0000

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
1 (1→2)	(15)	0.1168	-0.1225	-0.1119	0.1272
1 (1→2)	(17)	0.1216	-0.1853	-0.1132	0.1934
1 (1→2)	(18)	0.1117	-0.0177	-0.1095	0.0198
1 (1→2)	(19)	0.1117	-0.0177	-0.1095	0.0198
2 (1→3)	(14)	0.2923	0.1203	-0.2774	-0.1060
2 (1→3)	(15)	0.3509	0.0676	-0.3291	-0.0467
2 (1→3)	(17)	0.3509	0.0370	-0.3297	-0.0166
2 (1→3)	(18)	0.3521	0.1184	-0.3285	-0.0958
2 (1→3)	(19)	0.3521	0.1184	-0.3285	-0.0958
3 (2→3)	(14)	0.2923	0.1203	-0.2774	-0.1060
3 (2→3)	(15)	0.2418	0.1808	-0.2262	-0.1658
3 (2→3)	(17)	0.2431	0.2126	-0.2256	-0.1959
3 (2→3)	(18)	0.2394	0.1288	-0.2264	-0.1163
3 (2→3)	(19)	0.2394	0.1288	-0.2264	-0.1163

Caso 5busfrank_ctap.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
1 (1→2)	(14)	-0.0000	-0.0000	0.0000	0.0000
1 (1→2)	(15)	0.1168	-0.1225	-0.1119	0.1272
1 (1→2)	(17)	0.1216	-0.1853	-0.1132	0.1934
1 (1→2)	(18)	0.1117	-0.0177	-0.1095	0.0198
1 (1→2)	(19)	0.1117	-0.0177	-0.1095	0.0198
2 (1→3)	(14)	0.2923	0.1203	-0.2774	-0.1060
2 (1→3)	(15)	0.3509	0.0676	-0.3291	-0.0467
2 (1→3)	(17)	0.3509	0.0370	-0.3297	-0.0166
2 (1→3)	(18)	0.3521	0.1184	-0.3285	-0.0958
2 (1→3)	(19)	0.3521	0.1184	-0.3285	-0.0958
3 (2→3)	(14)	0.2923	0.1203	-0.2774	-0.1060
3 (2→3)	(15)	0.2418	0.1808	-0.2262	-0.1658
3 (2→3)	(17)	0.2431	0.2126	-0.2256	-0.1959
3 (2→3)	(18)	0.2394	0.1288	-0.2264	-0.1163
3 (2→3)	(19)	0.2394	0.1288	-0.2264	-0.1163

Caso 9bus.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
1 (7→8)	(14)	0.5423	0.0149	-0.5401	-0.1753
1 (7→8)	(15)	0.4980	-0.0078	-0.4961	-0.1481
1 (7→8)	(17)	0.4982	0.0162	-0.4961	-0.1466
1 (7→8)	(18)	0.4982	0.0162	-0.4961	-0.1466
1 (7→8)	(19)	0.4982	0.0162	-0.4961	-0.1466
2 (4→5)	(14)	0.9220	0.0884	-0.9146	-0.2334
2 (4→5)	(15)	0.8596	0.0993	-0.8528	-0.2398
2 (4→5)	(17)	0.8619	0.1532	-0.8539	-0.2560
2 (4→5)	(18)	0.8619	0.1532	-0.8539	-0.2560
2 (4→5)	(19)	0.8619	0.1532	-0.8539	-0.2560
3 (4→6)	(14)	0.6545	-0.0865	-0.6485	-0.0692

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
3 (4→6)	(15)	0.5653	-0.0923	-0.5606	-0.0624
3 (4→6)	(17)	0.5673	-0.0490	-0.5618	-0.0772
3 (4→6)	(18)	0.5673	-0.0490	-0.5618	-0.0772
3 (4→6)	(19)	0.5673	-0.0490	-0.5618	-0.0772
4 (6→9)	(14)	-0.2515	-0.2308	0.2536	-0.1882
4 (6→9)	(15)	-0.3394	-0.2376	0.3434	-0.1577
4 (6→9)	(17)	-0.3382	-0.2228	0.3429	-0.1147
4 (6→9)	(18)	-0.3382	-0.2228	0.3429	-0.1147
4 (6→9)	(19)	-0.3382	-0.2228	0.3429	-0.1147
5 (7→5)	(14)	0.3387	-0.0787	-0.3354	-0.2666
5 (7→5)	(15)	0.4020	-0.0620	-0.3972	-0.2602
5 (7→5)	(17)	0.4018	-0.0266	-0.3961	-0.2440
5 (7→5)	(18)	0.4018	-0.0266	-0.3961	-0.2440
5 (7→5)	(19)	0.4018	-0.0266	-0.3961	-0.2440
6 (8→9)	(14)	-0.4599	-0.1747	0.4620	-0.0576
6 (8→9)	(15)	-0.5039	-0.2019	0.5066	-0.0172
6 (8→9)	(17)	-0.5039	-0.2034	0.5071	0.0203
6 (8→9)	(18)	-0.5039	-0.2034	0.5071	0.0203
6 (8→9)	(19)	-0.5039	-0.2034	0.5071	0.0203
7 (1→4)	(14)	1.5765	0.1210	-1.5765	-0.0020
7 (1→4)	(15)	1.4249	0.1088	-1.4249	-0.0070
7 (1→4)	(17)	1.4292	0.2226	-1.4292	-0.1042
7 (1→4)	(18)	1.4292	0.2226	-1.4292	-0.1042
7 (1→4)	(19)	1.4292	0.2226	-1.4292	-0.1042
8 (2→7)	(14)	0.8810	-0.0235	-0.8810	0.0638
8 (2→7)	(15)	0.9000	-0.0260	-0.9000	0.0698
8 (2→7)	(17)	0.9000	0.0399	-0.9000	0.0103
8 (2→7)	(18)	0.9000	0.0399	-0.9000	0.0103
8 (2→7)	(19)	0.9000	0.0399	-0.9000	0.0103
9 (3→9)	(14)	0.7157	-0.2181	-0.7157	0.2458
9 (3→9)	(15)	0.8500	-0.1374	-0.8500	0.1749
9 (3→9)	(17)	0.8500	-0.0524	-0.8500	0.0943
9 (3→9)	(18)	0.8500	-0.0524	-0.8500	0.0943
9 (3→9)	(19)	0.8500	-0.0524	-0.8500	0.0943

Caso 9bus_transformer_fields.pwf

Scripts equivalentes a (14)/(15) (mesmo cluster): (16).

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
1 (7→8)	(14)	0.5386	-0.0148	-0.5366	-0.1479
1 (7→8)	(15)	0.4980	-0.0078	-0.4961	-0.1481
1 (7→8)	(17)	0.4982	0.0047	-0.4961	-0.1404
1 (7→8)	(18)	0.4974	0.0305	-0.4953	-0.1683
1 (7→8)	(19)	0.4974	0.0305	-0.4953	-0.1683
2 (4→5)	(14)	0.9199	0.0787	-0.9126	-0.2252
2 (4→5)	(15)	0.8596	0.0993	-0.8528	-0.2398
2 (4→5)	(17)	0.8612	0.1777	-0.8534	-0.2911
2 (4→5)	(18)	0.8600	0.2181	-0.8523	-0.3385
2 (4→5)	(19)	0.8600	0.2181	-0.8523	-0.3385
3 (4→6)	(14)	0.6486	-0.1136	-0.6426	-0.0439
3 (4→6)	(15)	0.5653	-0.0923	-0.5606	-0.0624
3 (4→6)	(17)	0.5666	-0.0292	-0.5613	-0.1059
3 (4→6)	(18)	0.5673	0.0280	-0.5621	-0.1687

Ramo (f→t)	Script	pf (pu)	qf (pu)	pt (pu)	qt (pu)
3 (4→6)	(19)	0.5673	0.0280	-0.5621	-0.1687
4 (6→9)	(14)	-0.2574	-0.2561	0.2596	-0.1673
4 (6→9)	(15)	-0.3394	-0.2376	0.3434	-0.1577
4 (6→9)	(17)	-0.3387	-0.1941	0.3430	-0.1585
4 (6→9)	(18)	-0.3379	-0.1313	0.3423	-0.2277
4 (6→9)	(19)	-0.3379	-0.1313	0.3423	-0.2277
5 (7→5)	(14)	0.3408	-0.0725	-0.3374	-0.2748
5 (7→5)	(15)	0.4020	-0.0620	-0.3972	-0.2602
5 (7→5)	(17)	0.4018	-0.0755	-0.3966	-0.2089
5 (7→5)	(18)	0.4026	-0.1316	-0.3977	-0.1615
5 (7→5)	(19)	0.4026	-0.1316	-0.3977	-0.1615
6 (8→9)	(14)	-0.4634	-0.2021	0.4656	-0.0330
6 (8→9)	(15)	-0.5039	-0.2019	0.5066	-0.0172
6 (8→9)	(17)	-0.5039	-0.2096	0.5070	0.0189
6 (8→9)	(18)	-0.5047	-0.1817	0.5077	-0.0116
6 (8→9)	(19)	-0.5047	-0.1817	0.5077	-0.0116
7 (1→4)	(14)	1.5684	0.0825	-1.5684	0.0349
7 (1→4)	(15)	1.4249	0.1088	-1.4249	-0.0070
7 (1→4)	(17)	1.4278	0.2618	-1.4278	-0.1486
7 (1→4)	(18)	1.4273	0.3571	-1.4273	-0.2461
7 (1→4)	(19)	1.4273	0.3571	-1.4273	-0.2461
8 (2→7)	(14)	0.8795	-0.0472	-0.8795	0.0873
8 (2→7)	(15)	0.9000	-0.0260	-0.9000	0.0698
8 (2→7)	(17)	0.9000	-0.0217	-0.9000	0.0707
8 (2→7)	(18)	0.9000	-0.0524	-0.9000	0.1010
8 (2→7)	(19)	0.9000	-0.0524	-0.9000	0.1010
9 (3→9)	(14)	0.7252	-0.1733	-0.7252	0.2003
9 (3→9)	(15)	0.8500	-0.1374	-0.8500	0.1749
9 (3→9)	(17)	0.8500	-0.0985	-0.8500	0.1397
9 (3→9)	(18)	0.8500	-0.1963	-0.8500	0.2393
9 (3→9)	(19)	0.8500	-0.1963	-0.8500	0.2393

Caso test_defaults.pwf

Sem referência convergida; análise pulada.

Caso test_line_shunt.pwf

Sem referência convergida; análise pulada.

Caso test_system.pwf

Sem referência convergida; análise pulada.

4. Avaliação de Impacto Sistêmico

Tabela de perdas totais (em pu, na base do caso). $\Delta P_{loss\%}$ calculado em relação a (14) e (15) quando ambas convergiram. Para converter para MW, multiplique pela baseMVA do caso (tipicamente 100 MVA).

Caso	(14)	(15)	(16)	(17)	(18)	(19)	$\Delta\%$ vs(14): 17/18/19	$\Delta\%$ vs(15): 17/18/19
01 MAXIMA NOTURNA_DEZ25.PWF	—	—	—	—	—	—	— / — / —	— / — / —
300bus.pwf	—	5.77785	5.66572	5.67328	5.93561	5.93797	— / — / —	-1.81% / +2.73% / +2.77%
3bus.pwf	0.00753	0.00890	0.00890	0.00928	0.00928	0.00928	+23.25% / +23.25% / +23.25%	+4.24% / +4.24% / +4.24%
3bus_DBSH.pwf	0.02819	0.02919	0.02919	0.03014	0.03014	0.03014	+6.90% / +6.90% / +6.90%	+3.27% / +3.27% / +3.27%
3bus_DCER.pwf	0.02819	0.02919	0.02919	0.03032	0.03032	0.03032	+7.55% / +7.55% / +7.55%	+3.90% / +3.90% / +3.90%
3bus_DCSC.pwf	0.00006	0.00575	0.00271	0.00288	0.00288	0.00288	+4692.92% / +4692.92% / +4692.92%	-49.87% / -49.87% / -49.87%
3bus_DCline0.pwf	0.00753	—	6.13444	6.13444	6.13444	6.13444	+81387.72% / +81387.72% / +81387.72%	— / — / —
3bus_DSHL.pwf	—	1.03045	0.17739	0.17742	0.17742	0.17742	— / — / —	-82.78% / -82.78% / -82.78%
3bus_corrections.pwf	—	—	—	—	—	—	— / — / —	— / — / —
3bus_shunt_010229.pwf	0.01040	0.02624	0.02190	0.02190	0.02190	0.02190	-13.42% / -13.42% / -13.42%	+110.44% / +110.44% / +110.44%
3busfrank.pwf	0.02076	0.02708	0.02708	0.02655	0.02655	0.02655	+27.89% / +27.89% / +27.89%	-1.94% / -1.94% / -1.94%
3busfrank_continuous_shunt.pwf	0.01816	0.01889	0.01889	0.01817	0.01817	0.01817	+12.42% / +12.42% / +12.42%	-3.82% / -3.82% / -3.82%
3busfrank_q1023.pwf	0.16015	0.14665	0.14162	0.14162	0.14162	0.14162	+14.61% / +14.61% / +14.61%	-11.57% / -11.57% / -11.57%
4busfrank_vlim.pwf	—	—	0.14588	0.14574	0.14574	0.17144	— / — / —	— / — / —
500bus.pwf	—	0.86293	0.86366	0.90431	0.90431	0.90432	— / — / —	+4.80% / +4.80% / +4.80%

Caso	(14)	(15)	(16)	(17)	(18)	(19)	$\Delta\%$ vs(14): 17/18/19	$\Delta\%$ vs(15): 17/18/19
5busfrank.pwf	0.02404	0.03130	0.03130	0.02953	0.02953	0.02953	+22.83% / +22.83% / +22.83%	-5.65% / -5.65% / -5.65%
5busfrank_cp.pwf	0.03297	0.04591	0.04591	0.05071	0.05071	0.05071	+53.78% / +53.78% / +53.78%	+10.46% / +10.46% / +10.46%
5busfrank_csc.pwf	—	—	0.22963	0.17393	0.17393	0.17393	— / — / —	— / — / —
5busfrank_cp.pwf	0.03297	0.04591	0.04591	0.05071	0.04196	0.04197	+53.78% / +27.27% / +27.27%	+10.46% / -8.58% / -8.58%
5busfrank_cp.pwf	0.03297	0.04591	0.04591	0.05071	0.04196	0.04197	+53.78% / +27.27% / +27.27%	+10.46% / -8.58% / -8.58%
9bus.pwf	0.02316	0.02491	0.02491	0.02917	0.02917	0.02917	+25.96% / +25.96% / +25.96%	+17.08% / +17.08% / +17.08%
9bus_transformerfield.pwf	0.02491	0.02491	0.02491	0.02777	0.02732	0.02732	+20.25% / +18.33% / +18.32%	+11.46% / +9.68% / +9.68%
test_defaults.pwf	—	—	—	—	—	—	— / — / —	— / — / —
test_line_shunt.pwf	—	—	—	—	—	—	— / — / —	— / — / —
test_system.pwf	—	—	0.00229	0.00243	0.00243	0.00243	— / — / —	— / — / —

5. Conclusão Técnica

Robustez de convergência (sobre 25 casos):

- (14): 15/25 (60.0%)
- (15): 17/25 (68.0%)
- (16): 21/25 (84.0%)
- (17): 21/25 (84.0%)
- (18): 21/25 (84.0%)
- (19): 21/25 (84.0%)

Interpretação física dos controles:

- **QLIM** (limites de Q dos geradores; introduzido em (16)): impõe o setpoint de tensão das barras PV via slack quadrático `sl_v[i]`. Quando o reativo de máquina satura, a barra migra efetivamente para PQ — visualizável em casos onde `vm_pu` em (16)/(17)/(18) diverge do PF tradicional (15) porque a referência de tensão é abandonada pelo gerador saturado. Esperado: pequena perturbação no perfil de tensão em barras vizinhas a geradores com folga reduzida de Q.
- **CSCA** (controle de susceptância shunt; introduzido em (17)): libera `bs_var[i]` como variável de decisão dentro de `[bsmin, bsmax]` lido do bloco DOPC do PWF. Onde o ANAREDE permite chaveamento (banco

capacitivo/reativo), o OPF pode redespachar reativo localmente — tende a aliviar tensão em barras de carga e reduzir geração reativa de máquinas, podendo diminuir perdas. Esperado: ΔP_{loss} negativo (perdas menores) em casos com shunts controláveis significativos.

- **CTAP** (controle de tap em transformadores; introduzido em (18)): libera `tm_var[1]` como contínuo em `[tapmin, tapmax]`. Modifica a relação de transformação para redistribuir fluxos e tensões secundárias. Diferente do CSCA (que adiciona Q local), o CTAP redistribui fluxo ao longo do ramo. Pode competir com o CSCA — ambos atuam sobre tensão, mas com mecanismos físicos distintos. Esperado: efeito de segunda ordem no perfil de tensão e nas perdas; impacto maior em redes com folga de tap.
- **DERA** (corte de carga preventivo; introduzido em (19)): adiciona `corte_p[1]` `[0, pd]` e `corte_q[1]` (vinculado por fator de potência da carga). Penalidade hierárquica por nível de tensão ($1e7$ para <69 kV, $5e7$ para $69\text{--}230$ kV, $1e8$ para >230 kV) faz com que o solver prefira cortar carga na distribuição antes de cortar em EAT. Esperado: corte zero quando a rede está OK; corte positivo apenas em casos onde os outros controles não bastam para satisfazer as restrições.

Interações:

- *CTAP* $\times$ *CSCA*: podem atuar sobre o mesmo objetivo de tensão, mas o solver tende a preferir a manobra de menor “custo” ($\text{PENALIDADE_MENOR} = 1e4$ para shunt, sem penalidade para tap) — ou seja, CTAP age primeiro quando disponível.
- *DERA* $\times$ *resto*: DERA tem peso $1e7$ ($\gg \text{PENALIDADE} = 1e6$ dos slacks). Logo, o solver só cortará carga quando o resto realmente não conseguir respeitar QLIM/VLIM dentro das margens disponíveis. Casos onde DERA mostra corte > 0 (visíveis pela diferença de carga entre (19) e (14)/(15) na tabela de perdas) indicam estresse efetivo da rede.

Convergência: quanto mais variáveis e slacks o modelo adiciona, mais sensível à inicialização e às tolerâncias o Ipopt fica. Casos pequenos (3bus/5bus) convergem facilmente em todas as formulações; redes maiores (300bus, 500bus, SIN) podem expor diferenças de robustez entre (14)/(15) (PowerModels) e as variantes JuMP (16)–(19). Iter count não foi capturado nesta bateria (Ipopt MOI não expõe `BarrierIterations` para os solves rodados); para análise futura, pode-se parsear o log do Ipopt onde aparece “Number of Iterations.....”.